Wildlife Information System

(WIS)

Software Requirements and Use Cases

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1. Functional Requirements

Each line below says one thing the system must do. .The IDs are there so we can point to them during testing.

1.1 Login and Access Control

ID	Requirement
FR-01	The system shall let a user log in with email and password.
FR-02	The system shall return a JWT that holds the user'.s role and expires after a set time.
FR-03	The system shall show each role only its own menu items and pages.
FR-04	The system shall block API calls that have no valid token or API key (HTTP 401) and calls the role is not allowed to make (HTTP 403).
FR-05	The system shall let a logged-in user switch to a different role view only if they have that role assigned.
FR-06	The system shall store passwords as bcrypt hashes with a salt.

1.2 Animal Profiles

ID	Requirement
FR-07	The Admin shall be able to add, edit and delete animal records.
FR-08	Each animal shall get a unique ID automatically, made from the species and a number, for example COW-001 or TIGER-001.
FR-09	An animal record shall hold species, name, breed, gender, age, weight, zone and tag ID.
FR-10	The system shall check all fields before saving and return clear field errors if something is missing or wrong.
FR-11	The animals page shall list all animals in a table with ID, species, name, zone, health, activity, temperature and heart rate.
FR-12	The animals table . shall be filterable by species.
FR-13	Clicking an animal in the table shall open its details on the dashboard.
FR-14	The top search box shall .. find an animal by ID, name or species.

1.3 Live Map and Simulation

ID	Requirement
FR-15	The dashboard shall show a satellite map with the zones drawn as coloured rectangles with names.
FR-16	Each animal .. shall appear on the map as an icon with its ID label and a coloured dot for its health status.
FR-17	Hovering over an animal shall show a short card with species, zone, health, activity, temperature and heart rate.
FR-18	Clicking an animal shall select it and show its details in the side panel.
FR-19	The map shall support zoom in, zoom out, fit all animals and fullscreen.
FR-20	The simulation engine shall move each animal toward a random target point inside its own zone and never outside it.

ID	Requirement
FR-21	The simulation engine shall change each animal's behavour from time to time (for example grazing, walking, resting, drinking, stalking) using behavours that suit the species.
FR-22	The simulation engine shall keep temperature between 37.2 and 40.5 C and heart rate between 45 and 120 bpm, changng them a little at each update.
FR-23	The system shall update animal data every 10 seconds by default. The interval shall be changeable between 5 and 60 secnds.
FR-24	The dashboard shall also show recent activity, a 24 hour activity chart and a zone overvie w with animal counts.

1.4 Zones

ID	Requirement
FR-25	The Admin shall be able to add a zone with a name, color and boundary (north, south, east and west values).
FR-26	The Admin shall be able to draw the zone boundary directly on the map instead of typing values.
FR-27	New zones shall apear on the map, in the zone overview and in zone drop-down lists.

1.5 Movement and History

ID	Requirement
FR-28	The system shall save each animal's position with a timestamp at regular steps.
FR-29	The animal panel shall have a History ta b with the distance travelled, number of points, a small route map and an activity timeline.
FR-30	A user shall be able to replay an animal's route on the map and stop the replay.
FR-31	A user shall be able to show the animal's trail on the map for a short time.
FR-32	A user shall be able to download an animal.s history as a CSV file.
FR-33	Data Entry through CSV upload shall remain available. The system shall check evry row, save valid rows, skip bad rows and show a report of what was skipped and why.

1.6 Health and Veterinary Records

ID	Requirement
FR-34	The system shall show each animal's temperature, heart rate, hlth score and last vet check.
FR-35	The system shall show temperature and heart rate trend charts for the selected animal.
FR-36	The system shall set the health status automatically: Healthy up to 39.5C, Warning above 39.5C, Critical above 39.9 C.
FR-37	A Veternarian or Admin shall be able to add and edit vaccination, diagnosis and treatment records. Each record needs a date and a short summary.
FR-38	The Records tab shall list vaccination and medical records with date and status.
FR-39	The system shall keep DNA sample records linked to each animal (sample ID, collection date, lab name, result and species confirmed flag).

1.7 Devices and Sensors

ID	Requirement
FR-40	The syste m shall list devices (GPS collars, health sensors, camera traps, environmental sensors) with ID, linked animal, type, battery, signal, status and last seen time.
FR-41	The page shall show totals for all devices, online devices, devices with warnings and devices with low battery.
FR-42	An Admin or Field Officer shall be able to add a device and link it to an animal.
FR-43	A device shall be marked Low Battery when its battery falls to 20 percent or less.

1.8 Incidents, Alerts and Notifications

ID	Requirement
FR-44	A user with access shall be able to report an incident with a type, related animal, zone and description.
FR-45	Incident types shall include injured animal, anmal missing, abnormal behaviour, human-wildlife conflict, sensor failure and environmental hazard.
FR-46	The incidents page shall list incidents with status Open or Resolved and show totals for this month.
FR-47	A user shall be able to open an incident, read its details and mark it as resolved.
FR-48	The system shall create a notification when an animal's health status becomes Warning or Critical or when a device goes offline.
FR-49	The notification bell shall show the number of unread items and a list of the latest ones, with a Mark all read option.

1.9 Field Tasks and Patrol

ID	Requirement
FR-50	A Field Officer shall see a list of today's tasks with title, zone, pri.ority, due time and done status.
FR-51	A Field Officer shall be able to add a task and tick or untick it as done. Totals for completed and pending tasks shall update at once.
FR-52	The field tasks page shall show an urgent alert card for the most serious open alert, with a Respond button.
FR-53	A Field Officer shall be able to view the day's patrol route on a map with start, checkpoints and end point.
FR-54	A Field Officer shall be able to change the patrol route by clicking points on the map, and start the patrol.

1.10 Reports and Analytics

ID	Requirement
FR-55	The reports page shall show total animals, healthy animals, average temperature and average heart rate.
FR-56	The reports page shall show charts for species distribution, health distribution and animals per zone.
FR-57	The report numbers shall refresh while the .simulation is running.

1.11 Search, Export and API Access

ID	Requirement
FR-58	The system shall s earch animals by species, location or date range and show results in pages of 20.
FR-59	Researchers shall be able to download movement data, health records, activity log, a complete dataset and an analytics summary as CSV, and a full export as JSON.
FR-60	The researchr page shall show the user's API key with Copy and Regenerate buttons.
FR-61	The researcher page shall have a data explorer where the user picks an endpoint, adds query parameters and sees the JSON or CSV response.
FR-62	The system shall show API usage: requests today, rate limit, data points available and the time of the last export.
FR-63	The API shall return JSON, use standard HTTP status codes and require a JWT or API key on every request except login.
FR-64	The system shall log every API request made with an API key.

1.12 User Management and Settings

ID	Requirement
FR-65	The Admin shall be able to add, edit, deactivate and delete users and assign one of the roles.
FR-66	The system shall make sure e ach user has a unique email address.
FR-67	The users page shall show name, email, role, status and last login, along with the role permission table.
FR-68	The Admin shall be able to start and pause the simulation, set the update interval, switch animal movement, health simulation and alerts on or off, and set the movement speed (slow, normal, fast).
FR-69	The Admin shall be able to reset the simulation to its starting data and export all data.
FR-70	The Admin shall be able to see and generate API keys and see recent webhook events.

2. Non-Functional Requirements

ID	Type	Requirement
NFR-01	Performance	Standard API requests shall respond within 2 seconds.
NFR-02	Performance	The live map shall keep moving smoothly with at least 50 animals on screen.
NFR-03	Scalability	The database shall handle up to 1 million ..animal and movement records without noticeable slowdown.
NFR-04	Security	All API calls shall use HTTPS and require JWT or API key authentication.
NFR-05	Security	Passwords shall be hashed with b crypt and salt. API keys shall be shown in full only when created.
NFR-06	Availability	The system shall be available 99 percent of the time after deployment.
NFR-07	Data integrity	Every record shall have created and updated timestamps, and the system shall log who created or changed it.
NFR-08	Usability	A new user shall be able to use the main pages for their role without training.

ID	Type	Requirement
NFR-09	Compatibility	The web app shall work in current Chrome, Firefox and Edge, on desktop and tablet screens.
NFR-10	Compatibility	The API shall follow REST rules so any language or tool can use it.

3. Use Cases

3.1 Actors

Actor	Description
Admin	Runs the whole system and manages users, data and settings.
Field Officer	Works in the field and uses tasks, patrols and incidents.
Researcher	Studies the data and exports it.
Veterinarian	Looks after health and vaccination records.
External App (API user)	Another program that reads data using an API key.
Simulation Engine	Background service on the server that creates live animal data. It has no screen.

3.2 Use Case Summary

ID	Name	Main actor	Category
UC-01	Login and Logout	All users	Access
UC-02	View Live Map and Animal Details	Admin, Field Officer, Researcher, Vet	Monitoring
UC-03	Manage Animal Records	Admin	Data management
UC-04	View Movement History and Replay Route	Admin, Field Officer, Researcher	Monitoring
UC-05	Manage Zones	Admin	Data management
UC-06	Control Simulation	Admin	Simulation
UC-07	Manage Health Records	Veterinarian, Admin	Data management
UC-08	Manage Devices and Sensors	Admin, Field Officer	Devices
UC-09	Report and Resolve Incident	Field Officer, Admin	Field work
UC-10	Manage Field Tasks	Field Officer	Field work
UC-11	Plan and Start Patrol Route	Field Officer	Field work
UC-12	View Reports and Analytics	Researcher, Vet, Admin	Reports
UC-13	Export Data	Researcher, Vet, Admin	Reports
UC-14	Access Data Through API	Researcher, External App	API
UC-15	Manage Users and Roles	Admin	User management
UC-16	View Notifications and Alerts	All logged-in users	Monitoring

3.3 Use Case Diagram

This diagram shows every role and what it can do. Admin could do everything the others can, but to keep the picture readable Admin is only linked to the admin tasks.



Figure 2: Use case diagram for WIS

3.4 Use Case Diagrams by Role

Admin

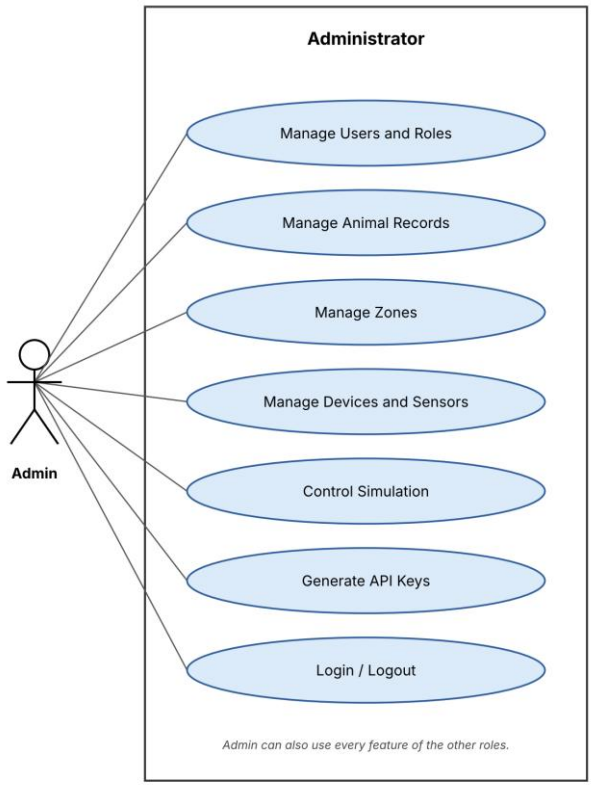


Figure 3: Admin use cases

Field Officer

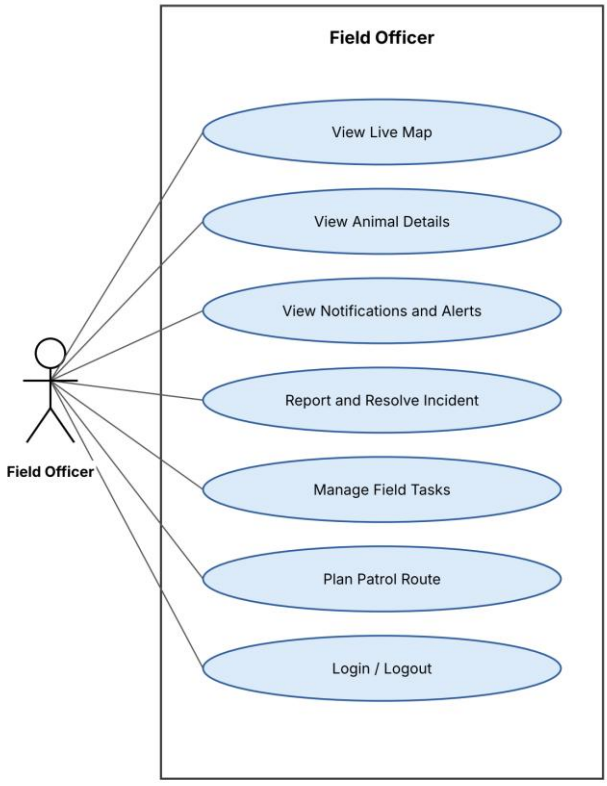


Figure 4: Field Officer use cases

Researcher and API User

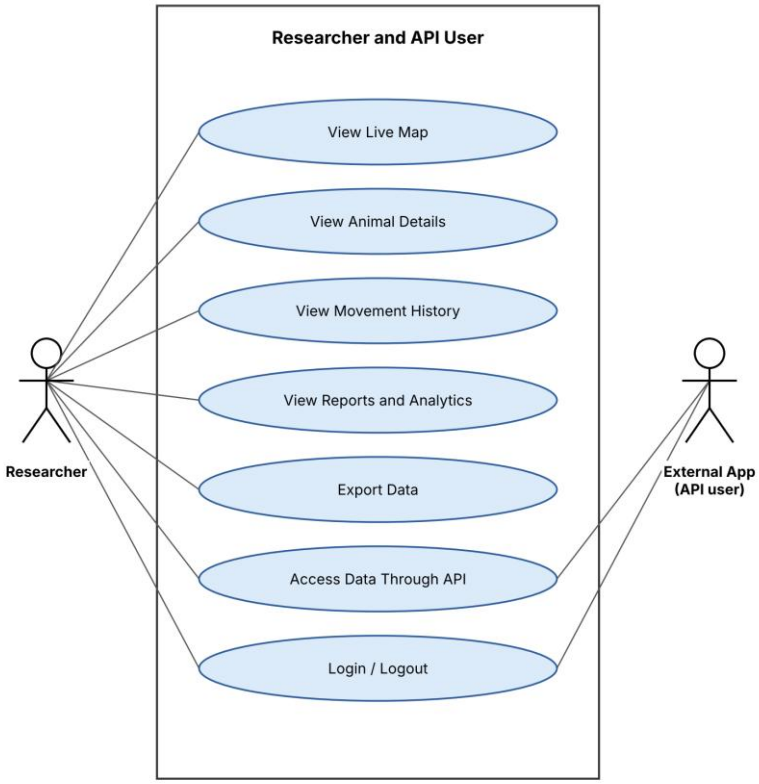


Figure 5: Researcher and API user use cases

Veterinarian

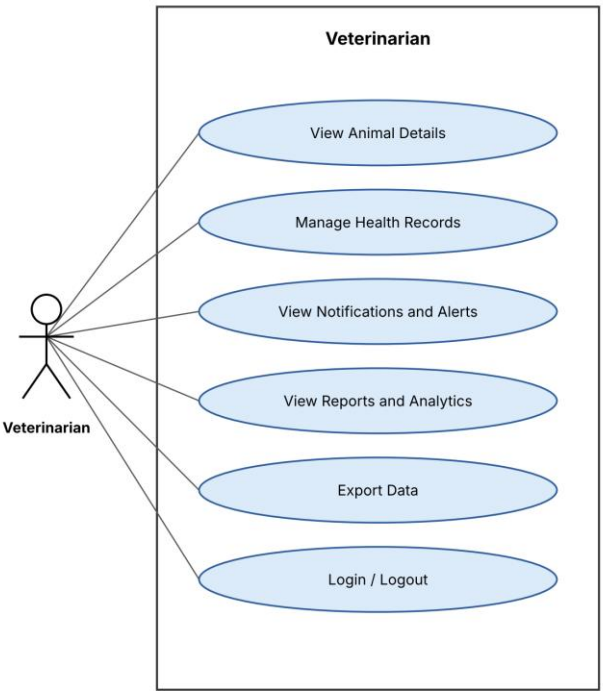


Figure 6: Veterinarian use cases

3.5 Use Case Descriptions

UC-01: Login and Logout

Actor	All users
Pre-condition	The user has an account and it is active.
Post-condition	The user is logged in and sees the menu and default page for their role. On logout the token is removed.
Main Flow	1. User opens the login page and enters email and password. 2. System finds the user by email and compares the password with the bcrypt hash. 3. System creates a JWT that holds the user's role. 4. System op ens the default page for the role (Admin: Dashboard, Researcher: Researcher API, Field Officer: Field Tasks).
Alternate Flow	Wrong email or password: HTTP 401 with a general message that doesn't say which one was wrong. Deactivated account: HTTP 403.

UC-02: View Live Map and Animal Details

Actor	Admin, Field Officer, Researcher, Veterinarian
Pre-condition	User is logged in.
Post-condition	The user sees animals on the map and the full details of the chosen animal.
Main Flow	1. User opens the Dashboard or Map page. 2. System shows the map, zones and animals at their current positions. 3. User hovers .over an animal to see the short info card. 4. User clicks the animal. 5. System opens the side panel with Overview, Health, Records, Location and History tabs. 6. System keeps updating temperature, heart rate and position every update interval.
Alternate Flow	Map tiles fail to load: the system still shows animals and zones and shows a message. Animal not found: HTTP 404.

UC-03: Manage Animal Records

Actor	Admin
Pre-condition	User is logged in as Admin.
Post-condition	An animal is added, ..updated or deleted. A new animal has a unique ID.
Main Flow	1. Admin opens the Animals page and clicks Add Animal. 2. Admin enters species, name, breed, gender, age, weight and zone. 3. System checks all fields. 4. System creates the ID (for example COW-002), saves the animal and places it inside its zone. 5. System shows a success message and the animal appears on the map and in the table.
Alternate Flow	Missing or invalid fields: HTTP 400 with a message for each field. Deleting an animal asks for confirmation first.

UC-04: View Movement History and Replay Route

Actor	Admin, Field Officer, Researcher
Pre-condition	User is logged in. The animal has saved positions.
Post-condition	The user has seen the route, or downloaded the history as CSV.
Main Flow	1. User selects an animal and opens the History tab. 2. System shows distance, number of points, route map and activity timeline. 3. User clicks Play @ Route. 4. System moves a marker along the saved path. 5. User clicks Download to get the history as a CSV file.
Alternate Flow	Fewer than two saved points : system shows the message 'Not enough trail data'.

UC-05: Manage Zones

Actor	Admin
Pre-condition	User is logged in as Admin.
Post-condition	A new zone is saved and shown on all maps and drop-down lists.
Main Flow	1. Admin clicks Add Zone on the map. 2. Admin enters a name, color and boundary values, or draws the boundary on the map by clicking points and double-clicking to finish. 3. Admin saves the zone. 4. System checks that the name is filled and the boundary is valid. 5. System adds the zone to the map, zone overview and forms.
Alternate Flow	No name or fewer than 3 drawn points : system shows a warning and does not save. Admin can cancel drawing at any time.

UC-06: Control Simulation

Actor	Admin
Pre-condition	User is logged in as Admin.
Post-condition	The simulation runs with the new settings.
Main Flow	1. Admin opens Settings. 2. Admin switches the simulation on or off, picks the update interval and speed, and switches movement, health simulation and alerts. 3. System saves the settings and applies them to the simulation engine. 4. Admin can press @ Reset Simulation to go back to the starting animals.
Alternate Flow	Reset asks for confirmation. If the engine cannot be reached, system shows an error and keeps the old settings.

UC-07: Manage Health Records

Actor	Veterinarian, Admin
Pre-condition	User is logged in as Veterinarian or Admin. The animal exists.
Post-condition	A vaccination, diagnosis or treatment record is saved for the animal.
Main Flow	1. User opens the animal and goes to the Records tab. 2. User clicks Add Record and fills in type, date and a short summary. 3. System checks the fields. 4. System saves the record and shows it in the list with its status.
Alternate Flow	Missing date or summary: HTTP400. Animal ID not found: HTTP404. Other roles cannot add records: HTTP403.

UC-08: Manage Devices and Sensors

Actor	Admin, Field Officer
Pre-condition	User is logged in as Admin or Field Officer.
Post-condition	A device is added or its latest status is shown.
Main Flow	1. User opens Devices and Sensors. 2. System shows totals and the device table with battery and signal. 3. User clicks Add Device and enters device ID, type and animal. 4. System saves the device and shows it online.
Alternate Flow	Device ID already exists : HTTP 409. Battery 20 percent or lower: status shows Low Battery.

UC-09: Report and Resolve Incident

Actor	Field Officer, Admin
Pre-condition	User is logged in as Field Officer or Admin.
Post-condition	The incident is saved as Open, or changed to Resolved.
Main Flow	1. User opens Incidents and clicks Re port Incident. 2. User picks the type, animal and zone and writes a description. 3. System saves the incident as Open and adds it to the table. 4. Later, user opens the incident and clicks Resolve. 5. System sets the status to Resolved and updates the totals.
Alternate Flow	Empty description: system asks for one and does not save.

UC-10: Manage Field Tasks

Actor	Field Officer
Pre-condition	User is logged in as Field Officer.
Post-condition	Tasks are added or marked done, and the totals are updated.
Main Flow	1. Officer opens Field Tasks and sees the urgent alert, totals and task list. 2. Officer clicks Add Task and enters title, zone, priority and due time. 3. System saves the task. 4. Officer ticks a task when it is finished. 5. System updates the completed and pending counts.
Alternate Flow	Empty title:: system asks for a title. Officer can untick a task to reopen it.

UC-11: Plan and Start Patrol Route

Actor	Field Officer
Pre-condition	User is logged in as Field Officer.
Post-condition	The patrol route is saved and the patrol is marked active.
Main Flow	1. Officer opens Patrol Route and sees the current route with start, checkpoints and end. 2. Officer clicks Edit Route and clicks points on the map. A double-click finishes the route. 3. System draws and saves the ne.w route. 4. Officer clicks Start Patrol. 5. System marks the patrol as active.
Alternate Flow	Fewer than 2 points .: system asks for at least 2 points. Officer can cancel editing and keep the old route.

UC-12: View Reports and Analytics

Actor	Researcher, Veterinarian, Admin
Pre-condition	User is logged in with one of these roles.
Post-condition	The user has seen the current numbers and charts.
Main Flow	1. User opens Reports or Analytics. 2. System shows total animals, healthy count, average temperature and average heart rate. 3. System draws species, health and zone charts. 4. Numbers refresh while the simulation runs.
Alternate Flow	No animals in the system: charts are empty and totals show 0.

UC-13: Export Data

Actor	Researcher, Veterinarian, Admin
Pre-condition	User is logged in with one of these roles.
Post-condition	A CSV or JSON file is downloaded.
Main Flow	1. User opens the download area and picks a dataset (movement, health, activity, complete dataset, analytics or JSON). 2. System collects the data with the current filters. 3. System builds the file. 4. Browser downloads the file.
Alternate Flow	No matching records : system gives a file with only headers and tells the user.

UC-14: Access Data Through API

Actor	Researcher, External App
Pre-condition	The caller has a valid JWT or API key.
Post-condition	The caller gets a JSON (or CSV) response with status 200.
Main Flow	1. Caller sends a GET? request, for example /api/animals?species=Tiger, with the token or key. 2. System checks the token or key and the role. 3. System runs the query. 4. System returns the result as JSON. 5. System counts the request in the usage numbers.
Alternate Flow	Invalid or expired token or key : ?401. Role not allowed: ?403. Rate limit passed: 429. No results: ?200 with an empty list.

UC-15: Manage Users and Roles

Actor	Admin
Pre-condition	User is logged in as Admin.
Post-condition	A user is created, changed, deactivated or deleted.
Main Flow	1. Admin opens Users and Roles and clicks Add User. 2. Admin enters name, email and role. 3. System checks that the email is not used. 4. System saves the user and creates a password hash. 5. User appears in the table. Admin can also edit, deactivate or delete a user.
Alternate Flow	Duplicate email : “'Email already exists'. Deleting a user asks for confirmation.

UC-16: View Notifications and Alerts

Actor	All logged-in users
Pre-condition	User is logged in.
Post-condition	The user has read the notifications and the unread count is updated.
Main Flow	1. The simulation engine finds an animal with Warning or Critical health, or a device that went offline. 2. System creates a notification. 3. The bell shows the unread count. 4. User opens the bell & reads the list. 5. User clicks Mark all read.
Alternate Flow	No notifications : the list shows that nothing new has come in.